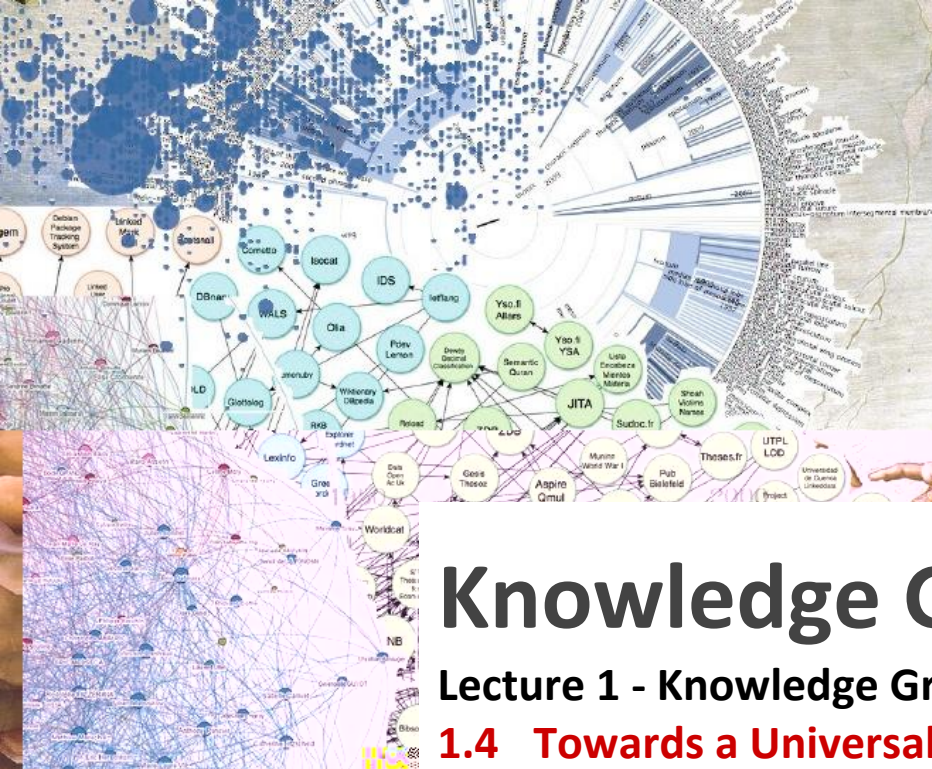




Knowledge Graphs

Lecture 1 - Knowledge Graphs in the Web of Data

1.4 Towards a Universal Knowledge Representation

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Autumn 2020



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Knowledge Graphs

Lecture 1: Knowledge Graphs in the Web of Data

1.1 Data, Information, and Knowledge

1.2 How to Represent Knowledge?

1.3 The Art of Understanding

1.4 Towards a Universal Knowledge Representation

1.5 The Semantic Web

1.6 Linked Data and the Web of Data

Climate Change is
the Everest of all



Knowledge and Understanding



Text: "

"

Entity Mapping
Entity Disambiguation

Everest, Kansas

a small village

Everest, Gasfield

a gas field near Scotland

George Everest

a Surveyor General of India

Jack Everest

an Irish football player

...

Mount Everest

a mountain

Disambiguation

solution of linguistic ambiguities

Knowledge and Understanding



The **Meaning (Semantics)**
of entities and classes must
be defined explicitly.

Text: "Everest"

Entity Resolution

Mount Everest

is a

Mountain

is a subclass of

Landform

is a subclass of

Geomorphical Unit

Entities

Classes

Knowledge and Understanding



`MountEverest` $\in$ `Mountain`

`Mountain` $\subseteq$ `Landform`

`Landform` $\subseteq$ `GeomorphicalUnit`

`GeomorphicalUnit` $\subseteq$ `NaturalGeographicObject`

`GeorgeEverest` $\in$ `Person`

`Person` $\cap$ `Mountain` = $\emptyset$

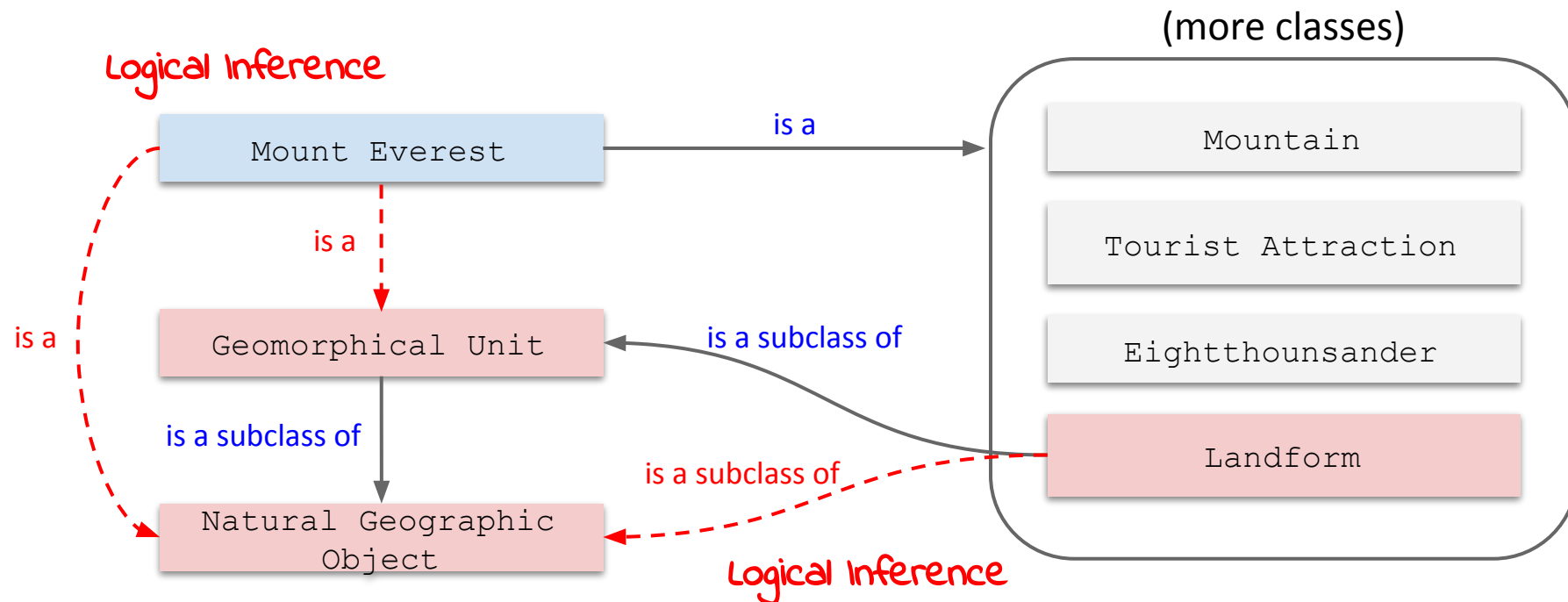
Logical Inference

`MountEverest` $\notin$ `Person`

`GeorgeEverest` $\notin$ `Mountain`

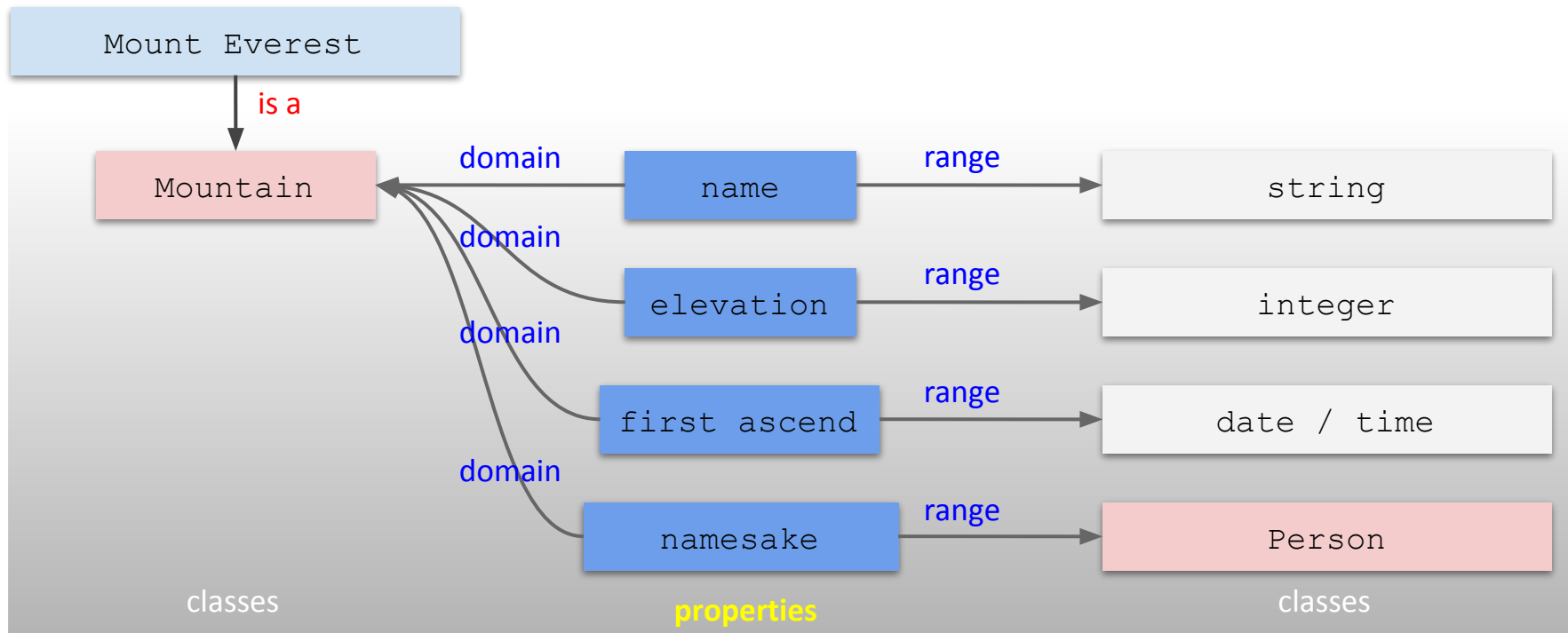
Knowledge and Understanding

The Meaning (Semantics) of information is expressed with the help of knowledge representations (**Ontologies**)



Knowledge and Understanding

The Meaning (Semantics) is expressed with the help of knowledge representations (**Ontologies**)



Knowledge Representation vs Data Structures

What's the difference to traditional data structures?

1. **Mathematical Logic** provides a framework to **formally express the semantics** of knowledge representations.
2. **Semantics** of knowledge representations can be defined **explicitly**.
3. **Mathematical Logic** enables **logical inferences** and **reasoning** for knowledge representations.

The Semantic Web - A Web of Data

The Semantic Web is an **Extension of the current Web**.

The meaning of information (Semantics) is made explicit by **formal (structured) and standardized knowledge representations (Ontologies)**.

Thereby it will be possible,

- to **process** the meaning of information automatically,
- to **relate** and **integrate** heterogeneous data,
- to **deduce** implicit (not evident) information from existing (evident) information in an automated way.

The Semantic Web is kind of a **global database** that contains a **universal network of semantic propositions**.

The Semantic Web - A Web of Data



„The Semantic Web is an extension of the current web in which information is given well-defined meaning, better enabling computers and people to work in cooperation.“

Tim Berners-Lee, James Hendler, Ora Lassila: [The Semantic Web](#), Scientific American, 284(5), pp. 34-43(2001)

The Semantic Web Technology Stack (not a piece of cake...)

Most apps use only a subset of the stack

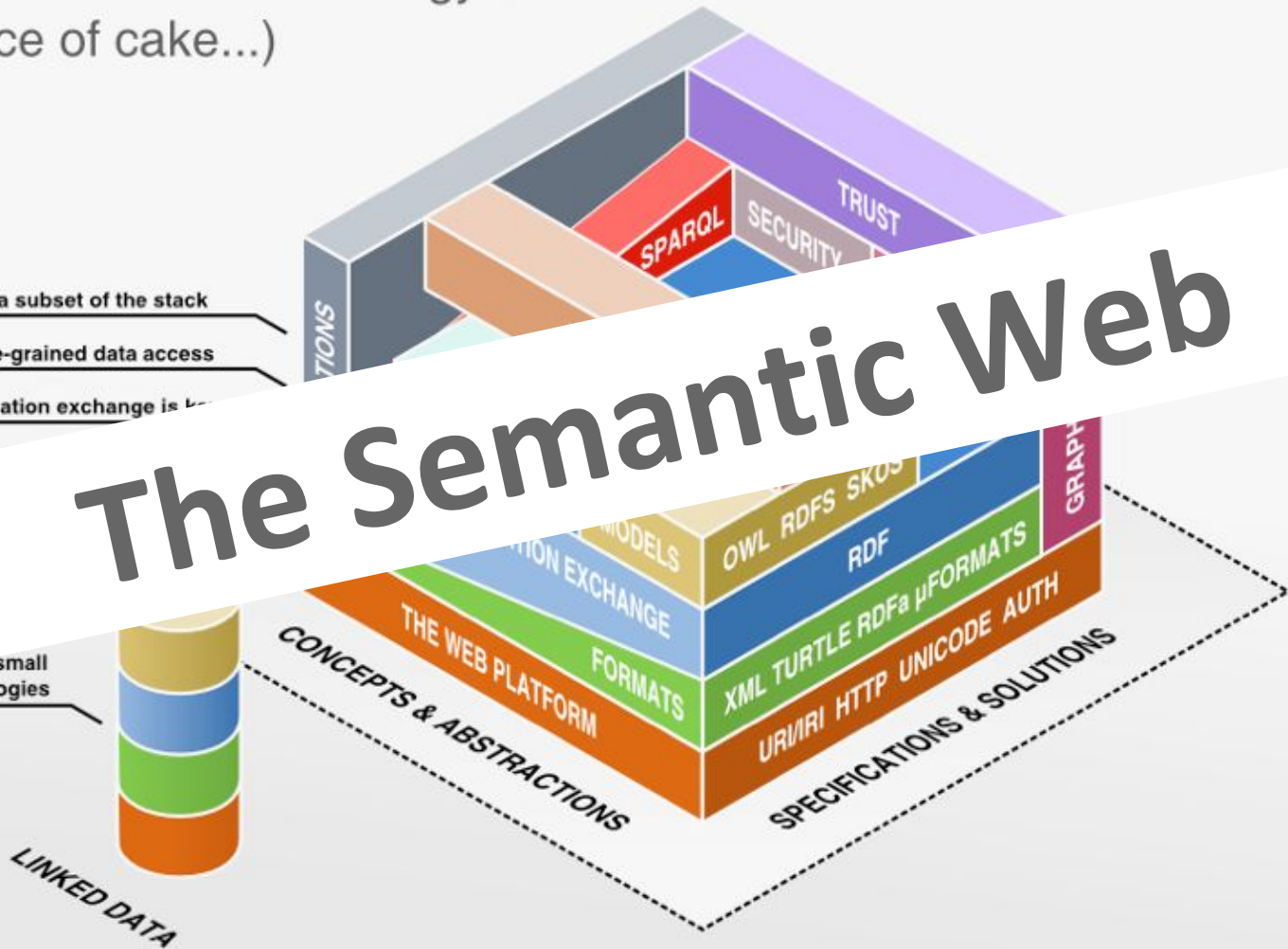
Querying allows fine-grained data access

Standardized information exchange is key

Formats are needed

The Semantic Web

Linked Data uses a small
selection of technologies



Next Lecture...

Picture References:

- [1] Mount Everest as seen from an aircraft from airline company Drukair in Bhutan.
Shrimpo1967 derivative work: Papa Lima Whiskey 2 (Diskussion) [CC BY-SA 2.0]
[https://commons.wikimedia.org/wiki/File:Mount Everest as seen from Drukair2 PLW edit.jpg?uselang=de](https://commons.wikimedia.org/wiki/File:Mount_Everest_as_seen_from_Drukair2_PLW_edit.jpg?uselang=de)
- [2] Tim Berners-Lee, James Hendler, Ora Lassila: [*The Semantic Web*](#), Scientific American, 284(5), pp. 34-43(2001)
- [3] Benjamin Nowack, *The Semantic Web - Not a Piece of cake...*, at bnode.org, 2009-07-08 , [CC BY 3.0]
<http://bnode.org/blog/2009/07/08/the-semantic-web-not-a-piece-of-cake>